

ACOUSTIC PROGRESS PROPAGATION FOR LONG-HORIZON SPECULATIVE DECODING IN ASR

Yuanyuan Jia Qianqian Yang

Zhejiang University, Hangzhou, China

ABSTRACT

Speculative decoding accelerates autoregressive automatic speech recognition (ASR), but the acceptance length of alignment-aware drafters can saturate as the draft horizon increases. We propose a progress-aware speculative drafter that recurrently propagates an acoustic progress state across draft steps and feeds it back into audio cross-attention to guide token generation. We jointly train the drafter and progress predictor over variable draft horizons. On five ASR test sets, our method achieves lossless, macro-averaged end-to-end speedups of $1.657\times$ and $1.227\times$ over target-only autoregressive decoding with Qwen3-ASR-0.6B and Qwen3-ASR-1.7B, respectively. Relative to AnchorDraft, our method improves the macro-averaged speedup by 34.3% and 9.0%, respectively. Horizon sweeps show continued growth in acceptance length beyond the baselines’ saturation. Code is available at <https://github.com/yuanyuanjia71-spec/ProgDraft>.

Index Terms— Automatic speech recognition, speculative decoding, inference acceleration

1. INTRODUCTION

Recent state-of-the-art automatic speech recognition (ASR) models typically employ an autoregressive Transformer-based decoder to generate transcriptions conditioned on acoustic representations [1, 2]. However, this autoregressive paradigm generates tokens one by one, resulting in high inference latency that limits its ability to meet the requirements of latency-sensitive applications, such as real-time transcription and voice assistants [3]. To overcome this limitation, speculative decoding [4, 5, 6, 7] has emerged as a possible solution. It uses a lightweight draft model to propose multiple candidate tokens, which the target model verifies in parallel within a single forward pass, reducing sequential decoding overhead.

Recent works have explored this approach for speech recognition. Whisper-Medusa [8] extends Medusa-style multi-head prediction [9] to ASR, while SpecASR [3] accelerates decoding through adaptive draft lengths and draft reuse. However, generating accurate ASR drafts also requires tracking which speech segment corresponds to each predicted

token. Because different tokens span different amounts of audio, advancing the text by one word or subword does not directly show where to look next in the speech. A lightweight drafter must therefore track its own acoustic position across multiple steps. Recent work [10] also shows that this audio attention often drifts away from the correct position during rollouts, even when the drafter can see the full audio context. This drift lowers token acceptance at later steps, limiting the speedup of longer drafts. To address this problem, AnchorDraft adds stepwise supervision to the drafter’s audio attention during training [10].

However, improving per-step alignment accuracy does not necessarily enable effective use of longer drafts. We observe pronounced long-horizon saturation. As draft length increases, AnchorDraft’s acceptance length quickly saturates, and additional draft steps yield little further decoding progress. Stepwise alignment supervision can improve the drafter’s acoustic localization accuracy, but it does not change how localization is performed at inference time. At each draft step, the drafter must use its current state to re-identify the acoustic region corresponding to the next token within the full acoustic context [10]. Thus, accurate local alignment over short horizons does not ensure that the drafter can continuously and reliably track its position in the audio as the draft trajectory advances.

Building on prior recurrent position updates and Gaussian acoustic-attention weighting [11], we propose a progress-aware speculative drafter that propagates acoustic progress from a target-attention anchor across draft steps to improve long-horizon acceptance. We further adopt variable-horizon training, randomly sampling draft lengths during training so that the model learns this acoustic progression over rollouts of different lengths, thereby improving stability and enabling more effective use of longer draft horizons.

Our contributions are threefold:

1. We propose acoustic progress propagation to guide successive speculative ASR draft steps through recurrent updates of an acoustic position state and feedback to audio cross-attention.
2. We jointly train the drafter and progress predictor over variable draft horizons, combining token prediction and

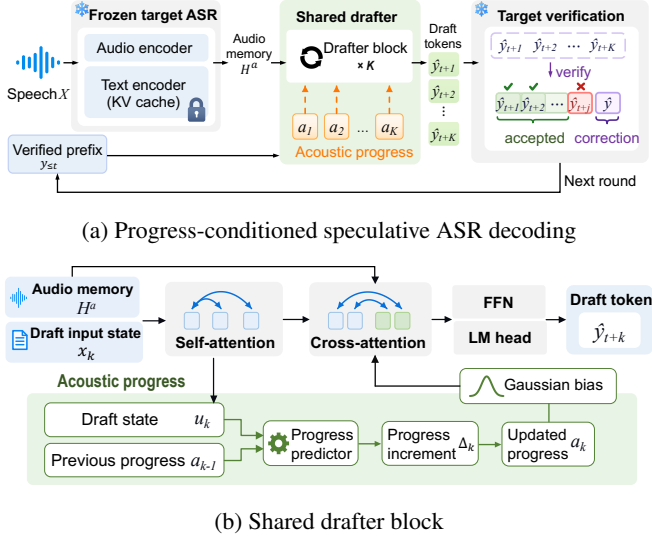


Fig. 1. Overview of the proposed progress-conditioned speculative ASR decoding. (a) Draft-and-verify pipeline. (b) Acoustic-progress prediction and feedback within one shared draft step.

alignment supervision to learn acoustic progress transitions across rollouts of different lengths.

3. Experiments on five ASR test sets show that our method makes more effective use of long draft horizons, achieving longer acceptance lengths and higher lossless end-to-end speedups than the evaluated baselines.

2. METHOD

2.1. Speculative Drafter Architecture

As illustrated in Fig. 1(a), given speech X encoded by the target model as H^a and a verified prefix $y_{\leq t}$, the drafter proposes K candidates $\hat{y} = (\hat{y}_{t+1}, \dots, \hat{y}_{t+K})$. The target verifies them in parallel, accepting the longest prefix consistent with its greedy predictions. Unless decoding terminates, it appends a correction token at the first mismatch or a bonus token if all candidates are accepted [4].

We use a lightweight acoustic-conditioned drafter initialized by fusing the embedding $E(y_t)$ of the last committed token with hidden features $F_\ell(t)$ from target decoder block ℓ after processing that token:

$$x_1 = W_f [E(y_t); F_{\ell_1}(t); \dots; F_{\ell_m}(t)], \quad (1)$$

where $[\cdot; \cdot]$ denotes concatenation, W_f is a linear projection, and $\ell_1, \dots, \ell_m$ index the m selected target decoder blocks. Multi-layer target-feature fusion is also used in EAGLE-3 [12]. Restart features are extracted during a single-token

target context update; both this pass and batched verification are included in timing (Sec. 3.1).

As shown in Fig. 1(b), a single Transformer decoder block [13] applies causal self-attention, cross-attention over the full H^a , and a feed-forward network to produce draft state z_k from input x_k . The frozen target LM head maps z_k to a token distribution, whose argmax gives $\hat{y}_{t+k}$. The next input combines the draft state with the new token embedding:

$$x_{k+1} = z_k + E(\hat{y}_{t+k}). \quad (2)$$

The target parameters remain frozen, and the drafter hidden size matches the target embedding and LM-head input dimensions. Draft states and the self-attention cache are carried across steps.

2.2. Acoustic Progress Propagation

The recurrence in Sec. 2.1 propagates hidden states and the self-attention cache but leaves acoustic progression to cross-attention. We condition audio cross-attention on a recurrently propagated acoustic progress state a_k , which estimates the temporal position in the encoded speech at step k .

Following prior work [10], we initialize each round from target layer 21 (zero-indexed), using the query at the last committed prefix token. Let $\mathcal{A}$ denote valid audio keys, s_j their timestamps, and $\alpha_{h,j}^{\text{tar}}$ the attention weight from head h . Averaging over all N_h heads, we select the peak as the initial anchor:

$$j^* = \arg \max_{j \in \mathcal{A}} \frac{1}{N_h} \sum_{h=1}^{N_h} \alpha_{h,j}^{\text{tar}}, \quad a_1 = s_{j^*}. \quad (3)$$

These weights are reused from normal target decoding without an extra forward pass. Later steps use drafter-predicted progress without further target-attention queries.

For $k \geq 2$, let u_k denote the representation after self-attention and before audio cross-attention. Following the monotonic position-update principle [11], we model acoustic progression as a recurrent state transition conditioned on u_k and the preceding progress state:

$$\begin{aligned} \Delta_k &= \text{Softplus}(f_\theta([a_{k-1}/T; u_k])), \\ a_k &= a_{k-1} + \Delta_k, \end{aligned} \quad (4)$$

where f_θ is the progress predictor and T is the utterance duration. Positions and increments are measured in seconds; only the predictor's position input is normalized by T .

Acoustic retrieval uses $\bar{a}_k$, the position clipped to $[0, T]$, to form a Gaussian log-bias added to cross-attention scores before softmax:

$$\begin{aligned} B_{k,j} &= [\Psi(\bar{a}_k)]_j = -\frac{(s_j - \bar{a}_k)^2}{2\sigma^2}, \\ S'_{k,j} &= S_{k,j} + B_{k,j}, \end{aligned} \quad (5)$$

Table 1. End-to-end speedup over target-only autoregressive decoding and average acceptance length τ on five ASR test sets with Qwen3-ASR-0.6B and Qwen3-ASR-1.7B. All methods use inference draft length $K = 8$.

Model	Method	LibriSpeech test-clean		LibriSpeech test-other		TED-LIUM 3 test		GigaSpeech test		FLEURS en-us test	
		Speedup	τ	Speedup	τ	Speedup	τ	Speedup	τ	Speedup	τ
Qwen3-ASR 0.6B	Base Drafter	1.203×	3.043	1.172×	2.976	1.227×	3.102	1.186×	3.012	1.131×	2.864
	SpeechSpec-EAGLE3	0.555×	1.545	0.560×	1.544	0.580×	1.616	0.573×	1.568	0.532×	1.451
	AnchorDraft	1.260×	3.241	1.223×	3.137	1.263×	3.245	1.246×	3.178	1.176×	2.986
	AnchorDraft + RC	1.120×	3.071	1.105×	3.009	1.179×	3.173	1.120×	3.047	1.065×	2.854
	Ours	1.786×	5.055	1.624×	4.553	1.737×	4.867	1.633×	4.533	1.504×	4.135
Qwen3-ASR 1.7B	Base Drafter	0.994×	2.871	0.968×	2.781	1.009×	2.918	0.986×	2.839	0.929×	2.663
	SpeechSpec-EAGLE3	0.588×	1.625	0.594×	1.640	0.619×	1.721	0.610×	1.669	0.558×	1.522
	AnchorDraft	1.157×	3.349	1.101×	3.202	1.150×	3.335	1.134×	3.267	1.089×	3.100
	AnchorDraft + RC	1.059×	3.147	1.021×	3.032	1.090×	3.196	1.057×	3.101	0.997×	2.929
	Ours	1.289×	4.009	1.224×	3.752	1.291×	3.967	1.216×	3.751	1.116×	3.416

where $S_{k,j}$ is the original score and $\sigma > 0$ controls the temporal width. This bias favors positions near $\bar{a}_k$ while retaining access to the full H^a ; recurrence and supervision use the unclipped a_k .

Through the recurrence in Eq. (2), the retrieved acoustic context influences u_{k+1} and thus the next progress prediction, closing the feedback loop.

2.3. Training

With the target frozen, we jointly train the drafter and progress predictor on target-generated greedy trajectories. Teacher forcing supplies $E(y_{t+k})$ to the next step, while progress follows its own recurrence from the target-attention anchor a_1 , without forced-alignment resets.

We apply token cross-entropy $\mathcal{L}_{\text{tok},i,k}$ at each unrolled depth. Both auxiliary losses use Smooth L1 ($\beta = 1$). Once per rollout, feature matching $\mathcal{L}_{\text{feat},i}$ compares $z_{i,1}$ with the target’s final normalized decoder state at position t , averaging the loss over hidden dimensions.

We align the target-generated transcript with TorchAudio’s MMS forced aligner [14] and use each token’s aligned time-span midpoint as $a_{i,k}^*$. For $k \geq 2$, $\mathcal{L}_{\text{prog},i,k}$ compares the unclipped $a_{i,k}$ with $a_{i,k}^*$ in seconds, excluding tokens without valid alignments. Progress feedback also passes token-loss gradients to the predictor.

For each anchor i , we independently sample K_i and unroll $K_i^{\text{act}} = \min(K_i, R_i)$ steps, where R_i counts remaining target-trajectory tokens, including terminal EOS. Define $m_{i,k} = \mathbf{1}\{K_i^{\text{act}} \geq k\}$ and the empirical occurrence rate $q_k = N_{\text{ep}}^{-1} \sum_{i=1}^{N_{\text{ep}}} m_{i,k}$ from the epoch’s rollout plan of N_{ep} anchors. Unexecuted steps contribute zero loss. The per-

anchor objective is

$$\begin{aligned} \mathcal{L}_i = & \sum_{k=1}^{K_{\max}} \frac{m_{i,k} \gamma^{k-1}}{q_k} \mathcal{L}_{\text{tok},i,k} + \lambda_f \mathcal{L}_{\text{feat},i} \\ & + \lambda_p \sum_{k=2}^{K_{\max}} \frac{m_{i,k} \gamma^{k-1}}{q_k} \mathcal{L}_{\text{prog},i,k}, \end{aligned} \quad (6)$$

where γ controls loss decay across draft depths, and λ_f and λ_p weight feature matching and progress supervision, respectively. The training horizon range is specified in Sec. 3.1.

3. EXPERIMENTS

3.1. Experimental Setup

Datasets and Target Models. We freeze Qwen3-ASR-0.6B and Qwen3-ASR-1.7B [2] and train drafters on 3,933 English utterances from LibriSpeech clean/other [15], TED-LIUM 3 [16], GigaSpeech [17], and FLEURS [18]. All methods share 200 utterances per test split (1,000 total; FLEURS en-us).

Baselines. *Base Drafter* uses the single-layer recurrent block (Sec. 2.1) with unrestricted audio cross-attention and no alignment supervision, progress prediction, or progress-based bias. *Base Drafter* and *AnchorDraft* [10] train at fixed $K_{\text{train}} = 3$ unless specified. *Base Drafter* sums equally weighted token losses and uses $\lambda_f = 0.5$. *AnchorDraft* + *RC* reuses the *AnchorDraft* checkpoint; at inference, peak head-averaged target verification attention centers the next round’s acoustic window. *SpeechSpec-EAGLE3* [19, 12] uses native five-step training unrolling. Drafters share each target’s generated corpus.

Metrics. All test token sequences matched Target-only AR, preserving word/character error rates (WER/CER). Throughout, we report end-to-end speedup over Target-only AR and

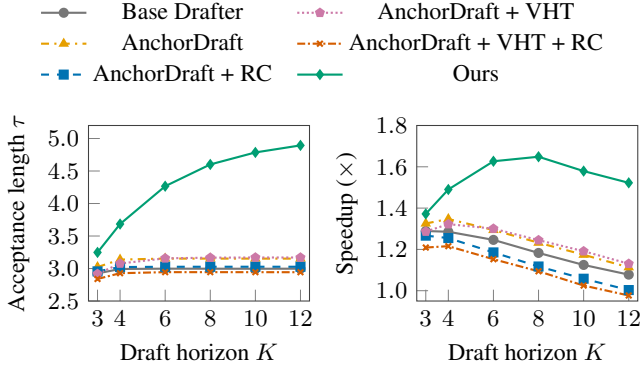


Fig. 2. Average acceptance length τ (left) and end-to-end speedup over Target-only AR (right) versus inference horizon K , with Qwen3-ASR-0.6B on the 1,000-utterance test set.

acceptance length $\tau = R^{-1} \sum_{r=1}^R (A_r + b_r)$, pooling all R rounds per evaluated set. A_r counts accepted draft-prefix tokens and $b_r \in \{0, 1\}$ the emitted correction/bonus token ($b_r = 0$ if EOS terminates without one).

Implementation. Except for the frozen-drafter ablation, drafters train for 45 epochs (14,760 updates) at effective batch size 12 for both targets. Our drafters contain 17.85M/71.34M parameters (0.6B/1.7B targets) and use features from decoder blocks 0, 9, 18, and 27 (zero-indexed). Our joint training uses AdamW [20] (learning rate 10^{-3} , cosine decay) and independent per-anchor sampling $K \sim \mathcal{U}\{3, \dots, 8\}$ (Sec. 2.3). We set $\sigma = 0.2$ s, $\gamma = 0.7$, $\lambda_f = 0.5$, and $\lambda_p = 0.1$. All methods decode greedily; end-to-end timing uses one H100 80GB GPU, batch size 1, FP32 eager attention for the target, and BF16 autocast for the drafter.

3.2. Main Results

Across all five test sets and both target model sizes, our method achieved the highest end-to-end speedup and average acceptance length (Table 1). Relative to AnchorDraft, the macro-averaged speedup and τ improved by 34.3% and 46.6%, respectively, with Qwen3-ASR-0.6B, and by 9.0% and 16.3% with Qwen3-ASR-1.7B. AnchorDraft consistently outperformed Base Drafter, but adding runtime correction reduced both metrics. SpeechSpec-EAGLE3 did not achieve end-to-end speedup under the shared-training-corpus setting.

Our drafter’s acceptance length continued to grow beyond baseline saturation (Fig. 2). Because verification accepts only a consecutive matching prefix, early mismatches limit the benefit of extending a draft. AnchorDraft with variable-horizon training (VHT; $K \sim \mathcal{U}\{3, \dots, 8\}$) plateaued by $K = 6$, with or without runtime correction, supporting acoustic progress propagation under matched training horizon ranges. Our method achieved the highest speedup at every tested horizon, peaking at $K = 8$. Beyond this point, further acceptance gains did not offset the increased cost of

longer draft-and-verify rounds.

3.3. Ablation Study

We evaluate six ablations (Table 2). *w/o Acoustic Progress* matches Base Drafter’s architecture and inference, training from scratch with our variable horizons and reweighted token loss (Eq. (6)). *w/o Progress Feedback* removes attention feedback but retains progress prediction and supervision. *Absolute-position prediction* uses $a_k = g_\phi(u_k)$ without position recurrence, retaining supervision and feedback. *w/o Joint Training* freezes the final Base Drafter. A randomly initialized predictor trains for five position-only epochs (selecting epoch 3), then five variable-horizon epochs with token CE and progress loss ($\lambda_p = 0.1$). We report the final checkpoint. *Fixed- K training* fixes $K = 3$ or $K = 8$.

Table 2. Ablations on Qwen3-ASR-0.6B using 1,000 test utterances and inference horizon $K = 8$.

Model Variant	Acceptance length $\tau \uparrow$	E2E Speedup $\uparrow$
w/o Acoustic Progress	3.078	1.206×
w/o Progress Feedback	3.069	1.210×
Absolute-position prediction	3.214	1.233×
w/o Joint Training	3.130	1.121×
Fixed- $K = 3$ training	3.760	1.331×
Fixed- $K = 8$ training	4.572	1.611×
Ours	4.601	1.649×

Table 2 supports the contribution of cross-step propagation: recurrence outperformed absolute-position prediction in both metrics with supervision and feedback retained, whereas supervision without feedback performed comparably to removing progress modeling. An explicit position reference may ease tracking compared with recovering absolute location from each hidden state.

The frozen-drafter variant underperformed joint training, but the schedules differ, so this comparison does not isolate joint optimization. Variable-horizon training outperformed fixed- $K = 8$ training in both metrics at the same maximum horizon, indicating gains beyond exposure to longer drafts. The fixed-horizon model also improved both metrics over $K = 3$.

4. CONCLUSION

We have proposed a speculative ASR drafter that propagates acoustic progress to guide audio cross-attention. Joint training over variable horizons improves its use of longer drafts. Across five ASR test sets and two target sizes, higher acceptance lengths and lossless end-to-end speedups over the evaluated baselines support acoustic progress propagation beyond local alignment alone.

5. REFERENCES

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